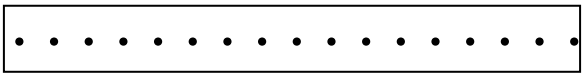
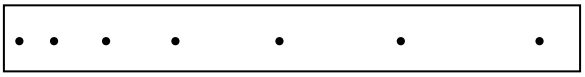
SCIENCE 1.6 **Name:** ***ANSWERS (A4Print)***

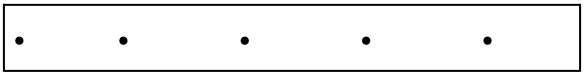
WORKSHEET TWO **ANALYSING TICKER TAPES**

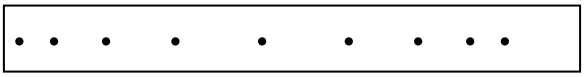
(Assume 50 dots per second and tapes are actual size when asked to do calculations)

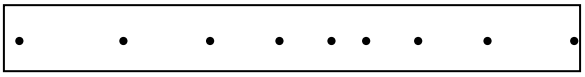
1. Describe the motion that produced the following ticker-tape portions.

(a)  ***slow constant speed***

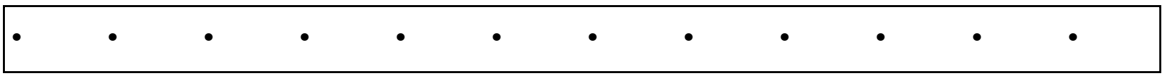
(b)  ***acceleration***

(c)  ***high constant speed***

(d)  ***accelerating then decelerating***

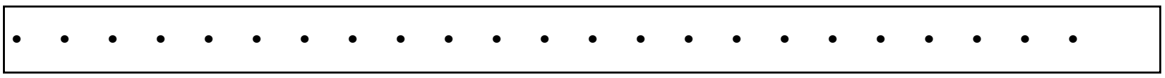
(e)  ***decelerating then accelerating***

2. Calculate the speed at which the following ticker tape is being pulled through the ticker timer.



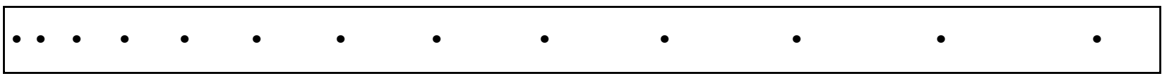
d = 14cm, t = 11 tics(0.22s) $v = \frac{d}{t} = \frac{14}{0.22}$ $= 64\ cms^{-1} = 0.64\ ms^{-1}$

3. Calculate the average speed of the journey recorded by the following ticker tape.



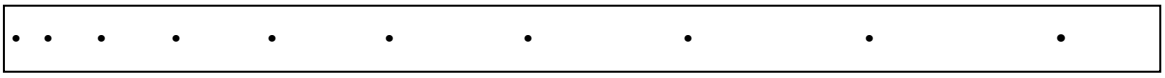
d = 14cm, t = 22 tics(0.44s) $v = \frac{d}{t} = \frac{14}{0.44}$ $= 32\ cms^{-1} = 0.32\ ms^{-1}$

4. Describe the motion shown in the following ticker tape ***acceleration***
Calculate the average speed of the journey.



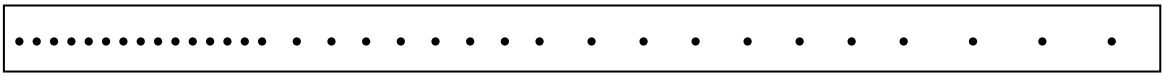
d = 14.3cm, t = 12 tics(0.24s) $v = \frac{d}{t} = \frac{14.3}{0.24}$ $= 60\ cms^{-1} = 0.60\ ms^{-1}$

5. Calculate the average speed of the journey represented by the following ticker tape.



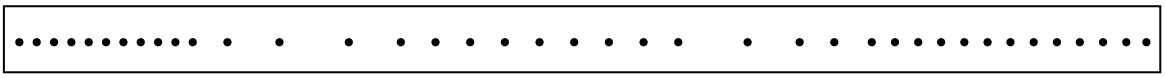
d = 13.9 cm, t = 9 tics(0.18s) $v = \frac{d}{t} = \frac{13.9}{0.18}$ *= 77 cms^{-1} = 0.77 ms^{-1}*

6. Calculate the average speed of the journey represented by the following ticker tape.



d = 14.5 cm, t = 32 tics(0.64s) $v = \frac{d}{t} = \frac{14.5}{0.64}$ *= 23 cms^{-1} = 0.23 ms^{-1}*

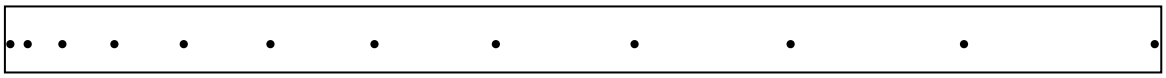
7. Calculate the average speed of the journey represented by the following ticker tape.



d = 15cm, t = 38 tics(0.76s) $v = \frac{d}{t} = \frac{15}{0.76}$ *= 20 cms^{-1} = 0.20 ms^{-1}*

Question 8 is for students who like a challenge! Calculate the acceleration of the ticker tapes.

(a)

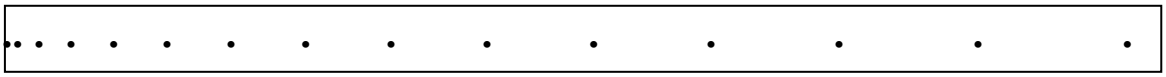


$v_1 = \frac{3.5}{0.1}$ = 35cms^{-1} (0.35ms^{-1}), $v_2 = \frac{10.3}{0.1}$ = 103cms^{-1} (1.03ms^{-1})

$\triangle t$ = 6tics (0.12s)

$$a = \frac{\Delta v}{\Delta t} = \frac{1.03 - 0.35}{0.12} = 5.7 \text{ ms}^{-2}$$

(b)



$v_1 = \frac{2.1}{0.1}$ = 21cms^{-1} (0.21ms^{-1}), $v_2 = \frac{8.5}{0.1}$ = 85cms^{-1} (0.85ms^{-1})

$\triangle t$ = 9tics (0.18s)

$$a = \frac{\Delta v}{\Delta t} = \frac{0.85 - 0.21}{0.18} = 3.6 \text{ ms}^{-2}$$